

Business Analytics Project Vision Document

A Customer Segmentation Analytics Initiative for First Bank of Nigeria Limited

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1. Business Vertical and Problem Context

1.1 Organizational Context

First Bank of Nigeria Limited (FirstBank), established in 1894 with its headquarters in Marina, Lagos, Nigeria, is the oldest and among the largest commercial banks in West Africa. As of 2024, the bank served over 42 million customer accounts, including digital wallets, through more than 820 business offices and an extensive agent banking network spanning 772 of Nigeria's 774 Local Government Areas (First Bank of Nigeria, 2024). This scale is both FirstBank's greatest asset and its central analytics challenge. A customer base spanning subsistence farmers, urban salary earners, SME owners, and high net worth clients cannot be served effectively through undifferentiated, one size fits all product offerings. Relationship managers currently rely heavily on broad demographic proxies such as age and branch location rather than behavioral and transactional patterns, resulting in generic cross sell offers and marketing spend that is not calibrated to how customers actually use their accounts.

FirstBank has publicly signaled a strategic shift toward the middle class consumer market and diversified retail portfolios, since consumer spending remains a comparatively small share of Nigeria's GDP relative to developed markets, representing significant untapped growth (Marketing Edge Magazine, 2024). Data driven segmentation directly supports this direction: it lets the bank convert an undifferentiated mass market customer base into an addressable set of segments it can serve profitably, while advancing financial inclusion, a stated strategic and regulatory priority in the Nigerian banking sector.

1.2 Core Process and Dataset

The core process is customer segmentation: grouping FirstBank's retail customers into behaviorally coherent segments using transaction history, account activity, and demographic attributes, to drive targeted cross sell offers, differentiated retention campaigns, and financial inclusion prioritization for underserved segments.

Attribute	Detail
Dataset	Bank Customer Segmentation (1M+ Transactions), Kaggle (Bansal, 2021)
Size and fields	Over 1 million anonymized transaction records linked to roughly 800,000 retail customers, well above the 5,000 record minimum. Includes customer ID, age, gender, location, balance, and transaction fields (Bansal, 2021).
Fit as proxy	FirstBank's proprietary core banking data is not accessible for an academic capstone. This public, real retail bank transaction dataset serves as a structurally realistic proxy for the same problem: demographics and transaction behavior mapped to segments. An equivalent internal dataset would come from FirstBank's core banking and card transaction systems.

2. Vision, Goals and Ethical Principles

2.1 Vision Statement

To equip FirstBank's retail banking and marketing teams with a data driven customer segmentation product that replaces broad demographic guesswork with evidence based behavioral segments, enabling more relevant product offers, more efficient marketing spend, and deliberate progress on financial inclusion, while holding the bank to an explicit standard of fairness, transparency, and accountability in how customer data is used.

2.2 Measurable Goals

- Increase cross sell conversion on targeted offers by at least 15% within two quarters of deployment, relative to the current undifferentiated campaign baseline.
- Identify at least three financially underserved segments and route them to a differentiated financial inclusion product track within one quarter.
- Maintain segment stability of at least 85% quarter over quarter and a documented fairness audit refreshed quarterly.

2.3 Ethical AI Vision and Fairness Objectives

FirstBank's segmentation product is built on the principle that customer data is a position of trust. This project commits to fairness (segments are evaluated for disparate impact before any commercial action), transparency (segment logic is explainable in business terms to relationship managers and customers on request), and accountability (every automated decision affecting a customer's product access has a named human owner and an audit trail).

- Ensure segment assignment does not disproportionately concentrate protected or historically underserved groups into lower value segments without a documented business justification.
- Review pricing and product recommendations for proxy discrimination, where a neutral variable such as location effectively substitutes for a protected attribute.
- Guarantee every customer facing decision informed by segmentation is contestable through a documented review process.

3. Users, Stakeholders and Ethical Oversight

3.1 Key User Groups

Key users include data analysts and data scientists who build and maintain the segmentation models, marketing and campaign managers who target retail campaigns, relationship managers and branch operations leads who use segment profiles in customer conversations, and product managers in the retail and financial inclusion units who use segment insights to shape new offerings.

3.2 Project Stakeholders

Stakeholder	Role	Interest
Head of Retail Banking	Executive sponsor	Revenue growth and portfolio diversification
Chief Marketing Officer	Business owner	Higher campaign return on investment
Chief Data and Analytics Officer	Technical sponsor	Data quality and model governance
Data Protection Officer	Compliance stakeholder	Alignment with Nigeria's data protection law
Customers	Ultimate beneficiaries	Fair treatment and protection of personal data

3.3 Ethics Oversight Body

Oversight sits with FirstBank's internal AI and Data Ethics Committee, chaired jointly by the Chief Data and Analytics Officer and the Data Protection Officer, with the Head of Retail Banking as a standing member. The committee reviews the Ethical Risk Register and fairness audit results before each major release and can pause deployment of any segment driven campaign pending further review.

4. Analytics Product and Technical Direction

4.1 Solution Overview

The deliverable combines predictive analytics (behavioral clustering and propensity scoring that predicts segment membership and likely product adoption) with prescriptive analytics (a recommendation engine that turns segment membership into a ranked list of next best offers). It is surfaced through a segment explorer dashboard, a customer lookup tool with plain language explanations, and a fairness monitor that tracks segment composition by protected attributes over time.

4.2 Proposed Technical Stack

Project Phase	Tool / Platform	Purpose
Ingestion and pipeline	AWS S3, AWS Glue, Apache Airflow, dbt	Land, catalog, clean, and transform data (Module 3)
Modeling and tracking	Python, scikit learn, AWS SageMaker, MLflow	Clustering, propensity modeling, fairness analysis (Module 4)
Dashboard	Power BI or Tableau	Segment explorer and fairness monitor for business users (Module 5)
Deployment	Docker on AWS ECS, API Gateway	Serve segment and propensity scores to the CRM
Version control	GitHub, GitHub Actions	Source control, review, and automated testing

4.3 Proposed Repository Structure

```
firstbank-customer-segmentation/  
  README.md  
  docs/      vision, ethics charter, data governance  
  data/      raw, interim, processed (DVC tracked)  
  notebooks/ EDA, feature engineering, clustering  
  src/  
    pipeline/ ETL and feature engineering scripts  
    models/   clustering and propensity model code  
    fairness/  bias and fairness audit scripts  
    api/      scoring service  
  dashboards/segment-explorer/  
  tests/  
  .github/workflows/ CI pipelines  
  requirements.txt  
  LICENSE
```

5. Requirements, Risks and Constraints

5.1 Requirements

- Functional: ingest demographic and transaction data on a scheduled batch basis and produce a segment label and propensity score for every active retail customer, refreshed monthly.
- Functional: provide role based dashboard access for marketing, retail, and compliance users, and expose a scoring API for the CRM and campaign system.
- Non functional: outputs must integrate with existing CRM tooling without custom middleware, access must be logged, and batch scoring must complete within a nightly processing window.

5.2 Ethical Risk Register

Risk	Description	Mitigation
Bias	Segments may correlate with protected attributes and lead to unequal offer quality	Quarterly fairness audit with ethics committee sign off before campaign launch
Privacy violation	Transaction level data increases exposure if mishandled or re identified	Pseudonymized identifiers, role based access, and encryption at rest and in transit
Misuse of segments	Segments could be repurposed for exclusionary pricing or predatory targeting	Explicit acceptable use policy that bans pricing discrimination use cases
Regulatory non compliance	Processing personal data without adequate legal basis under Nigeria's data protection law	Legal review of processing basis and alignment with NDPC registration requirements (National Information Technology Development Agency, 2023)

6. Quality, Documentation and Governance Standards

6.1 Documentation and Standards

Every module deliverable is version controlled in the GitHub repository shown in Section 4.3, with a corresponding entry in the docs folder explaining decisions made and why, so any reviewer can trace a business decision back to the data and code that produced it. This document follows Nexford University's professional reporting standards, including structured headings, NXU style in-text citations, and a complete reference list.

6.2 Versioning Strategy

- Code: Git Flow branching with required pull request review before merging to main.
- Data: DVC layered on top of Git to version raw and processed datasets.
- Models: MLflow registry with semantic versioning, where a major version change reflects a change in segmentation methodology.

6.3 Data Governance Principles

- Access: role based access control, with customer-level data limited to credentialed analytics and marketing staff.
- Usage and retention: data is used only for purposes documented in this charter, and raw extracts are retained only as long as needed for regulatory record keeping, consistent with Nigeria's data protection law (National Information Technology Development Agency, 2023).

References

1. Bansal, S. (2021). Bank customer segmentation (1M+ transactions) [Data set]. Kaggle. <https://www.kaggle.com/datasets/shivamb/bank-customer-segmentation>
2. First Bank of Nigeria. (2024). Home. <https://www.firstbanknigeria.com/home/>
3. Marketing Edge Magazine. (2024, June 8). FirstBank wins 2024 best SME in Nigeria, Africa. <https://marketingedge.com.ng/firstbank-wins-2024-best-sme-in-nigeria-africa/>
4. National Information Technology Development Agency. (2023). Nigeria Data Protection Act, 2023. Federal Republic of Nigeria.